# SUKALYAN ROY

Backend Engineer | Python, Data & ML Systems
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## SUMMARY

Backend engineer building data-intensive APIs, time-series storage, streaming paths, and model-serving interfaces in **Python, FastAPI, PostgreSQL/TimescaleDB, Redis, and Azure**. Comfortable owning a service from schema and API through deployment and measurement.

## PROFESSIONAL EXPERIENCE

**Backend Developer | Avinya Neurotech | Full-time, Remote | Jun 2025 to Present**

- Reworked a **five-service FastAPI platform**, exposing 50+ REST endpoints for EEG ingestion, analysis, export, and visualization.
- Set up and ran the **production and test server environments** on **Azure**: Docker Swarm services, PostgreSQL/TimescaleDB and Redis, **CI/CD releases**, versioned schema migrations, and **Prometheus/Grafana** observability.
- Designed the **TimescaleDB/PostgreSQL schema**, with COPY-based bulk ingestion and **time-windowed queries**, so live and historical reads over multichannel EEG come from the database rather than the source recordings.
- Built the **bounded WebSocket streaming path** with fixed-size binary frames and stale-request cancellation; tuning cut payload from **1.3 MB to 288 KB** per chunk and first-chunk latency from about 2.2 s to **783 ms p50**.
- Moved exports and frequency-band analysis off the request path with **Redis job queues** and Azure Blob Storage, then made recovery **ownership-safe** with leased jobs, heartbeats, and atomic requeue, so a restart neither loses nor duplicates work.
- Built the **sleep-analysis service** with a **memory-bounded chunked pipeline** that cut a 24 h recording from **157.6 s to 32.6 s** with identical results and completed a **50 h**, 32-channel study.
- Authored the device's **regulatory documentation**: the **CDSCO MD-12** test-licence dossier, **EU MDR** technical documentation, design history file, and IEC 62304 configuration records.
- Led a **signal-fidelity investigation** against a reference clinical viewer: matched traces to the raw EDF at **correlation above 0.999** and traced the reported drift to an **18.3%** display-scale difference, not a sample delay.

**Tech Development Intern | Aecho AI | Fully Remote | Nov 2024 to Jan 2025**

- Built a local-first RAG assistant over 300+ pages of internal documentation, cutting routine lookup from minutes to seconds while keeping documents and inference off hosted APIs.

## SELECTED PROJECTS

**MediPredict ML Microservice & ICIDA 2026 Paper | Dec 2025 to Aug 2026 | FastAPI, scikit-learn, SHAP, Docker**

- Sole author of the **FastAPI model-serving service**: typed schemas, artifact-driven model loading, preprocessing shared with training, health and readiness endpoints, SHAP payloads, and Docker packaging; a separate Spring Boot/React application built by the project team consumes it over HTTP.
- Traced a reported $R^2$ of **0.925** to **target leakage** (a realised-cost field unavailable at quote time) and showed by fixed-model feature ablation that the honest quote-time ceiling sits near $R^2$ 0.15.
- Cut the served artifact from **95.8 MB to 0.39 MB** and single-row inference from about **45 ms to 0.5 ms** at unchanged cross-validated accuracy.

**EEG Signal-Processing Benchmark & IEEE SPMB Submission | Jun 2026 to Oct 2026 | Python, NumPy, SciPy, MNE**

- Built a **reproducible runtime benchmark** for random-access EEG filtering on **53 validated recordings** from five public datasets, separating library-call validity, fallback policy, and measured error tolerance; revision submitted to **IEEE SPMB 2026**.
- Showed that two seconds of real context per side cuts display-seam error from **3.80–23.53 µV to below 0.001 µV**, at a worst-recording p95 DSP cost of **2.37 ms**.

**Local-First RAG Chatbot with MCP Integration | May 2026 to Aug 2026 | Python, FAISS, SQLite, Ollama**

- Built a **hybrid retrieval pipeline** fusing FAISS dense search with BM25 by reciprocal rank fusion, with SQLite keeping retrieval metadata inspectable and a JSON-RPC MCP server serving Streamlit and CLI clients.
- On a fixed **18-query synthetic regression set**, hybrid retrieval held Hit Rate@3 at 1.000 and moved MRR from **0.972 to 1.000**; regression evidence, not a general retrieval benchmark.

## TECHNICAL SKILLS

**Primary**: Python, SQL | **Working familiarity**: JavaScript, TypeScript, C
Backend and APIs: FastAPI, REST APIs, WebSockets, Node.js, asynchronous job processing
Data Systems: PostgreSQL, TimescaleDB, Redis, Azure Blob Storage, SQLite, FAISS
ML and Retrieval: PyTorch, scikit-learn, SHAP, RAG, Ollama, MCP, NumPy, pandas
Cloud and DevOps: Azure, Docker, Docker Swarm, GitHub Actions, CI/CD, Prometheus/Grafana, Linux/Bash, Pytest

## EDUCATION

**Bachelor of Technology in Computer Science and Engineering**
**Netaji Subhash Engineering College | Kolkata | Graduated 2026 | CGPA: 8.0/10.0**

## CERTIFICATIONS & ACHIEVEMENTS

**Smart India Hackathon National Finalist** (2023) | Google Project Management | Harvard CS50: CS50x, AI with Python, Cybersecurity